

Further Examples of Multicollinearity

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- The simplest way for two regressors to have (perfect) multicollinearity is when one regressor is a constant multiple of another (e.g., $X_1 = a \cdot X_2$, where X_1 and X_2 denote two regressors, and a is a scalar).
- For example, in estimating a relationship between consumption and income, it makes no sense to include as regressors income measured in dollars as well as income measured in thousands of dollars. One of these is redundant. See also Example 1 in page 5 of the lecture slides for Week 9.
- Remember the meaning of a coefficient in the multiple regression is the effect of this specific regressor after fixing the other regressors (after controlling for the other regressors).
- It does not make sense to hold income measured in dollars fixed while changing income measured in thousands of dollars.

- Another way that regressors can be perfectly collinear is when one regressor can be expressed as an **exact linear combination of the other regressors**. The **dummy variable trap** discussed in class is an example.
- However, note that the following example **does not have problem of (perfect) multicollinearity**:

$$cons = \beta_0 + \beta_1 inc + \beta_2 inc^2 + u,$$

where *cons* and *inc* refer to consumption and income, respectively. **This model is totally estimable**. This is because even though $x_2 = inc^2$ is an exact function of $x_1 = inc$, **inc^2 is not an exact linear function of inc** .

By contrast, if we replace inc^2 with $2 \cdot inc$, then we will have the multicollinearity problem.

Here are several usual examples of multicollinearity in empirical practice:

- Including the same regressor twice.
- Including a dummy variable and its square (note that $1^2 = 1$ and $0^2 = 0$).
- Estimating a regression on a sub-sample for which a dummy variable is either all zero or all one (see Example 2 in page 5 of the lecture slides for Week 9).